MIL-V-5027D
12 March 1968
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MIL-V-5027C
28 January 1966
MIL-V-8620A (Aer)
9 January 1959

MILITARY SPECIFICATION

VALVES, CHECK, OXYGEN, HIGH PRESSURE

This specification has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers high-pressure oxygen check valves.

1.2 Classification - The high-pressure oxygen check valves shall be of the following types, and of the following styles, as specified (see 6.2):

Type I - without spring-loaded check
Type II - with spring-loaded check

Type I, Style		An or MS Standard
A	Single check, tube-to-tube	AN6014
B (cone)	Dual check	AN6015
C	Dual check	AN6016
D	Dual check	AN6017
E	Cross check	AN6018
F	Single filler check, tube-to-tube	MS22035
G	Dual filler check	MS22035
Type II, Style		
B (flareless)	Dual check	MS90338

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 1660

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SPECIFICATIONS

Federal

O-T-236	Tetrachloroethylene (Perchloroethylene), Technical Grade
O-T-634	Trichloroethylene, Technical
NN-P-530	Plywood, Flat Panel
TT-W-572	Wood-Preservative, Water Repellent
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof-For Packaging and Sealing

Military

MIL-P-116	Preservation, Methods of
MIL-F-22191	Films, Transparent, Flexible, Heat Sealing for Packaging Applications
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, with Dispenser

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-794	Part and Equipment, Procedure for Packing and Packing of
MS22035	Valve, Check, Oxygen, High Pressure, Filler Connection

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MS33586	Metals, Definition of Dissimilar
MS90338	Valve, High Pressure Oxygen Check, Flareless Ends

Air Force-Navy Aeronautical

AN815	Union, Flared Tube
AN818	Nut - Coupling
AN6014	Valve - High Pressure Oxygen Check, Style A
AN6015	Valve - High Pressure Oxygen Check, Style B
AN6016	Valve - High Pressure Oxygen Check, Style C
AN6017	Valve - High Pressure Oxygen Check, Style D
AN6018	Valve - High Pressure Oxygen Check, Style E

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publication

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

American Society for Testing Materials Standard

D1149	Method of Test for Accelerated Ozone Cracking of Vulcanized Rubber

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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3. REQUIREMENTS

3.1 Qualification - The valves furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 First article - The valves furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Metal parts - All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.3.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.2 Nonmetallic materials - Any nonmetallic material that is adversely affected by continued use with oxygen shall not be used.

3.3.2.1 Age - Elastomer components, except silicone, shall not be more than 12 months old from the cure date of manufacture to the date of delivery to any Government Service or to any airframe or accessory manufacturer.

3.3.2.2 Elastomer components - The elastomer components shall be composed of an ozone resistant composition, which shall not bloom, and shall meet the specified ozone-resistance performance requirements.

3.3.3 Protective treatment - When materials are used in the construction of the valve that are subject to deterioration when exposed to environment conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance.

3.4 Design and construction - The design and construction of the valves shall be in accordance with the applicable AN or MS standards specified in the contract or order (see 6.2).

3.4.1 Types I and II -

3.4.1.1 Flow direction - Oxygen shall flow in the direction of the arrows as shown on the applicable drawings and shall be checked in the opposite direction.

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3.4.1.2 Gripping surfaces - If the valves are constructed in parts, each individual external part of the valves shall have a hexagonal surface or flats to serve as wrench-gripping surfaces. Adjacent to every thread of external connections, there shall be either a hexagonal surface or flats to serve as wrench-gripping surfaces.

3.4.1.3 Threads - Thread shall conform to Handbook H28.

3.4.1.4 Antiseize tape - Antiseize tape shall be used on all male pipe thread fittings. Antiseize tape shall conform to and shall be applied as specified in MIL-T-27730. Tape shall not be used on flare tube fitting straight threads, coupling sleeves, or on the outer side of tube flares. None of the tape shall be allowed to enter the inside of a fitting.

3.4.1.5 Degreasing - All internal parts of the valve shall be degreased using a vapor phase degreaser in accordance with O-T-236 or O-T-634. Ultrasonics may be used in conjunction with vapor phase degreasing for the cleaning of components. After completion of the cleaning and when assembled, General Electric Type H Leak Detector or equivalent Halide testing apparatus shall be used to determine the absence of the cleaning compound.

3.4.1.6 Orientation - The valve shall operate as specified herein when mounted in any position (see 4.6.4).

3.4.2 Type I -

3.4.2.1 Filter - An oxygen filter shall be installed in styles F and G to remove all foreign particles, larger than 65 microns, from the oxygen prior to passing through the check valve. Provisions shall be made for removal and replacement of the oxygen filter unit to facilitate cleaning.

3.4.3 Type II -

3.4.3.1 Poppet assembly - The valve shall incorporate a spring-loaded poppet or some other means to assure positive closing under all pressure and flow conditions when oriented in any position.

3.5 Performance -

3.5.1 Cracking pressure (type II) - The valve, when tested as specified in 4.7.2, shall require a pressure of at least 5 pounds per square inch gage (psig) but not more than 15 psig to open the valve check.

3.5.2 Reset pressure (type II) - The valve check, when tested as specified in 4.7.3, shall completely stop the flow of gas when the pressure is reduced to a range of from 4 psig to 14 psig.

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3.5.3 Flow check leakage -

3.5.3.1 Type I - The valve, when tested as specified in 4.7.4, shall not permit leakage in excess of 50 cc per minute.

3.5.3.2 Type II - The valve, when tested as specified in 4.7.4, shall not permit leakage in excess of 2 cc per minute.

3.5.4 Body leakage - The valve, when tested as specified in 4.7.5, shall not show any evidence of leakage.

3.5.5 Pressure drop -

3.5.5.1. Type I - The pressure drop across the valve, when tested as specified in 4.7.6, shall not exceed 15 psig.

3.5.5.2 Type II - The pressure drop across the valve, when tested as specified in 4.7.6, shall not exceed 22 psig.

3.5.6 High temperature - The valve, after being conditioned as specified in 4.7.7, shall pass the specified tests.

3.5.7 Low temperature - The valve, after being conditioned as specified in 4.7.8, shall pass the specified tests.

3.5.8 Vibration - The valve, when tested as specified in 4.7.9, shall not show any evidence of mechanical or material failure and shall pass the specified tests.

3.5.9 Endurance - The valve, when tested as specified in 4.7.10, shall pass the specified tests.

3.5.10 Simulated system charging (type II) - The valve, when tested as specified in 4.7.11, shall show no evidence of adverse effects due to testing and shall pass the specified tests.

3.5.11 Ozone-resistance - The test slabs, when tested as specified in 4.7.12, shall not show any evidence of checking, cracking, or damage.

3.5.12 Oxygen bomb - The test material and valve, when tested as specified in 4.7.13, shall not ignite and there shall not be any evidence of charring or deterioration.

3.5.13 Disassembly and examination - The valve, when examined as specified in 4.7.14, shall not show excessive evidence of deterioration or wear due to testing which might affect the valve performance.

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3.6 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Weight - The weight of each style valve shall be as specified in Table I.

TABLE I. Valve weights

Style	AN or MS standards	Weight maximum (ounces)
A	AN6014	6
B (cone)	AN6015	8
B (flareless)	MS90338	8
C	AN6016	7
D	AN6017	6
E	AN6018	8
F	MS22035	8
G	MS22035	9

3.8 Identification of product - The valves shall be marked for identification in accordance with MIL-STD-130, except that the Federal stock number shall be omitted.

3.9 Workmanship - The valves shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the valves shall be classified as follows:

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(a) Qualification inspection - Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products.

(b) First article inspection - First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item meets the requirements of this specification.

(c) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.3 Qualification inspection - Qualification inspection shall consist of all the examinations and tests of this specification.

4.3.1 Qualification samples - Qualification samples shall consist of four valves of each style and 6 elastomer test slabs, 6 inches by 6 inches by 0.075 inch, composed of the identical ozone-resistant composition and cure that was used in manufacturing the elastomer components. The qualification samples shall be forwarded to the Supply Officer, Naval Air Development Center, Johnsville, Warminister, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples submitted by (name) (date) for qualification inspection in accordance with requirements of MIL-V-5027D and number under authorization (reference authorizing letter and number) (see 6.4).

4.4 First article inspection - First article inspection shall consist of the following examinations and tests:

Visual examination
Cracking pressure (type II)
Reset pressure (type II)
Flow check leakage
Body leakage
Pressure drop

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4.4.1 First article samples -

4.4.1.1 Applicable to the Naval Air Systems Command - Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit 4 valves. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these valves from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity. Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article samples shall be forwarded to the Supply Officer, Naval Air Development Center, Johnsville Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department or to a laboratory designated by the procuring activity (see 6.2). The first article samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name) (date) for first article inspection in accordance with the requirements of MIL-V-5027D under contract no.

4.4.1.1.1 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the Government quality control representative. Two approved valves will be returned to the manufacturer for use in monitoring production. The other valves will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.4.1.2 Applicable to the Air Force and Army - For purchases by the Air Force and Army, the quantity and kind of samples, the name and address of the first article inspection activity and other pertinent information (see 4.4.1.1 and 4.4.1.1.1 for information) shall be as specified (see 6.2).

4.5 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Cracking pressure (type II)
Reset pressure (type II)
Flow check leakage
Body leakage
Pressure drop
Endurance
Simulated system charging (type II)
Preparation for delivery

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4.5.1 Sampling -

4.5.1.1 Inspection lot -

4.5.1.1.1 Valve - An inspection lot size shall be expressed in units of one valve made under essentially the same conditions and from the same materials and components. The sample unit shall be one valve.

4.5.1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing valves fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing valves fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of valves - The sample size, acceptance criteria, tests, and examinations required for the valves shall be as specified in Table II.

TABLE II

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATION OF THE VALVES

Inspection	Method	Sample Size	Acceptance Criteria
Visual examination (See classification of defects)	4.7.1.1	Every valve for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defects. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensions	4.7.1.1.1	Inspection Level S-2	An acceptable quality level of 1.0 defects per 100 units.
Cracking pressure (type II)	4.7.2	Every valve	Reject all defective valves.
Reset pressure (type II)	4.7.3	Every valve	Reject all defective valves.
Flow check leakage	4.7.4	Every valve	Reject all defective valves.

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TABLE II (Continued)

Inspection	Method	Sample Size	Acceptance Criteria
Body leakage	4.7.5	Every valve	Reject all defective units.
Pressure drop	4.7.6	Every valve	Reject all defective units.
Endurance 1/	4.7.10	Inspection Level S-2	Acceptance number zero, rejection number one.
Simulated system charging (type II) 1/	4.7.11	Inspection Level S-2	Acceptance number zero, rejection number one.
Examination of preparation for delivery	4.7.1.2	Inspection Level S-2	Total acceptable quality level of 4.0 percent defective.

1/ These are destructible test (See 6.2).

4.6 Test conditions -

4.6.1 Oxygen - Unless otherwise specified, the oxygen used in testing the valves shall conform to MIL-O-27210, type I.

4.6.2 Temperature and pressure - Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury and 70°F. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.6.3 For those valve styles that have more than one check, tests shall be applicable to each flow passage through the valve.

4.6.4 Orientation - When subjected to the flow check leakage and pressure drop tests, the valves shall be positioned with the centerline of the passage parallel to a vertical plane. Each consecutive valve tested shall be rotated 90 degrees about an axis perpendicular to the vertical plane from the position of the valve previously tested. The rotation shall always be in the same direction.

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4.7 Inspection methods -

4.7.1 Visual examination -

4.7.1.1 Valve - Every valve shall be examined visually (for critical defects) to determine conformance to this specification. The classification of defects, Table III, shall be used to classify the defects found.

4.7.1.1.1 Dimensions - The valve shall be checked dimensionally to determine conformance to the dimensions specified in applicable drawing.

TABLE III
CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE VALVE

Critical	Minor
1. Material imperfections - foreign matter embedded. 2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws. 3. Any component missing, malformed, fractured, or otherwise damaged. 4. Any component loose or otherwise not securely retained. 5. Incorrect assembling or improper positioning of components. 6. Any functioning part that works with difficulty. 7. Faulty workmanship or other irregularities.	201. Marking - missing, insufficient, incorrect, illegible, or not permanent.

4.7.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing valves selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, Table IV, shall be used to classify the defects found.

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TABLE IV
LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements.

4.7.2 Cracking pressure (type II) - A source of pressure shall be connected to the inlet of the valve. A pressure gage shall be connected to indicate the inlet pressure during the test. A flowmeter shall be connected to the outlet of the passage being tested. The inlet pressure shall be increased until a flow of 2 cubic centimeters per minute is indicated on the flowmeter. The inlet pressure at this rate of flow shall be recorded and shall pass the requirements specified in 3.5.1.

4.7.3 Reset pressure (type II) - The valve shall be set up for reset pressure testing as specified in 4.7.2. The inlet pressure shall be increased to a minimum of 50 psig. The pressure source shall then be cut off upstream from the pressure gage by a method such that the resetting of the check will trap gas in the valve and cause an indication of the pressure of the trapped gas on the pressure gage. The pressure of the trapped gas shall be recorded and shall pass the requirements specified in 3.5.2.

4.7.4 Flow check leakage - A source of pressure shall be connected to the outlet end or ends of the valve. Pressures of 50, 400, 1000 and 2000 psig shall be applied to the outlet end(s) and maintained for at least 1 minute at each pressure. The leakage shall be recorded and type I valve shall pass the requirements specified in 3.5.3.1 and the type II valve shall pass the requirements specified in 3.5.3.2.

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4.7.5 Body leakage - A source of pressure shall be connected to the inlet ends and the outlet ends of the valve sealed. A minimum pressure of 3,000 psig shall be applied to the inlet (s) for a minimum time of 2 minutes. The valve shall pass the requirements specified in 3.5.4.

4.7.6 Pressure drop - A constant pressure of 200 psig shall be applied to the inlet and a flow of 130 liters per minute maintained at the outlet. Under these conditions, the pressure drop across the valve shall be measured and the type I valve shall pass the requirements specified in 3.5.5.1 and the type II valve shall pass the requirements specified in 3.5.5.2.

4.7.7 High temperature - The valve shall be installed in a test chamber and the temperature shall be raised to 160 ±5°F and maintained for 3 hours. After the 3-hour stabilization period and while at this temperature, the type I valve shall be subjected to and pass the flow check leakage and pressure drop tests and the type II valve shall be subjected to and pass the cracking pressure, reset pressure, flow check leakage, and pressure drop tests.

4.7.8 Low temperature - The valve shall be installed in a test chamber and the temperature shall be lowered to -65 ±5°F and maintained for 3 hours. After the 3-hour stabilization period and while at this temperature, the type I valve shall be subjected to and pass the flow check leakage and pressure drop tests and the type II valve shall be subjected to and pass the cracking pressure, reset pressure, flow check leakage and pressure drop tests.

4.7.9 Vibration - A pressure of 2,000 psig shall be applied to the outlet end or ends of the valve. The valve shall then be continuously vibrated for a period of 6 hours, using a double amplitude of 0.018 to 0.020 inch at a varying frequency of 300 to 3,000 cpm. Every 2 hours during the vibration period and upon completion of the vibration period the type I valve shall be subjected to and pass the flow check leakage and pressure drop tests and the type II valve shall be subjected to and pass the cracking pressure, reset pressure, flow check leakage, and the pressure drop tests.

4.7.10 Endurance - The valve shall be connected to a 2,000-psi source of pressure. By a suitable cycling device or control valve, an intermittent flow of approximately 50 liters per minute shall be induced through the valve for approximately 3 seconds and then to zero flow for approximately 3 seconds. This will constitute one cycle. The test shall continue for a minimum of 20,000 cycles. The type I valve shall then be subjected to and pass the flow check leakage and pressure drop tests. The type II valve shall be subjected to and pass the cracking pressure, reset pressure, flow check leakage and the pressure drop tests. The supply pressure during this test may decay from 2,000 psig to 450 psig at which time the supply shall be recharged to 2000 psig.

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4.7.11 Simulated system charging (type II) - The inlet end of the valve shall be connected to a source of pressure with the outlet port uncapped. Valves that have more than one outlet port shall have additional outlets capped and be tested independently. By stopping the flow at the outlet port and then rapidly releasing the pressure, a chatter of the spring-loaded poppet may occur. Various flow rates through the outlet port may have to be used in order to produce this poppet chatter. When the chattering is produced, the flow rate through the outlet shall be maintained or slightly varied so that the chattering characteristics will continue. This valve chattering shall be maintained for 2 hours. The valve shall then be subjected to and pass the cracking pressure, reset pressure, flow check leakage and pressure drop tests.

4.7.12 Ozone-resistance (elastomer components) - Three of the six test slabs submitted for qualification shall be tested for ozone resistance. The test apparatus shall be in accordance with ASTM Method No. D1149. The test slabs shall be elongated 20 percent, placed in an ozone-free atmosphere for 24 hours, then placed in the ozone chamber. The chamber shall be adjusted to 100 ± 2°F and to give an exposure of ozone concentration of 120 ± 10 parts by volume of ozone per million parts by volume of air. The air-ozone velocity in the chamber shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test slabs shall then be examined under 10X magnification and shall pass the requirements specified in 3.5.11.

4.7.13 Oxygen bomb - Nonmetallic parts and the valve shall be subjected to oxygen bomb tests to determine compatibility of the items with high pressure oxygen.

4.7.13.1 Equipment - The equipment used in performing these tests shall be in accordance with Figures 1 and 2. The main valve shall be a rapid-opening 1/4 inch valve, shall require no internal lubrication for operation, and shall be rated for at least 3,000 psi operation. The valve shall be leak tight when closed and shall open with such rapidity that at least 95 percent of the total pressure rise shall have occurred within the test chamber in 20 ±7 milliseconds. The pressure rise time may be determined by measurement with a fast-responding small volume pressure transducer substituted for the test chamber in the test setup. The tubing and fitting connecting the valve to the test chamber shall be 3/16-inch size and shall be fabricated from corrosion-resistant steel. The run of tubing between the valve and the test chamber shall not be bent more than 4 degrees. The bleed valve shall be of a type which uses no organic material to accomplish its seal and requires no interior lubrication for operation. The bleed valve shall be installed so that when it is closed the test chamber does not communicate with the packing side of the valve. Test samples 3/16 inch diameter by 0.75 inch thick, shall be made from three of the test slabs submitted for qualification before the test slab is exposed to the ozone-resistance test.

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4.7.13.2 Procedure (elastomeric material) - The test samples shall be subjected to the following procedure: A sample shall be inserted in the corrosion-resistant steel recess of the test chamber and the test chamber assembled. The quick-acting valve shall be fully opened, and the test chamber pressurized with oxygen at 2175 ±25 psig for 30 seconds. At the end of the 30 second period, the pressure in the test chamber shall be released by means of the bleed valve, the test chamber opened and the sample shall be visually examined. If there is no indication of ignition, pressurizing operation shall be repeated at 60-second intervals. The pressurizing operations shall be repeated until either ignition occurs, or until three pressurizing operations have been performed. Remove the sample from the test chamber; the sample shall not show any evidence of charring or deterioration. With a fresh sample, the series of three pressurizing operations shall be repeated. This procedure shall be continued until either a sample has ignited, or until five test samples have been tested. The test material shall pass the requirements specified in 3.5.12.

4.7.13.3 Procedure (valve) - All connections on the valve except the high pressure inlet connection shall be sealed against a pressure of 2000 psig. The high pressure inlet connection of the valve shall be installed in the test setup on Figure 1 in place of the test chamber. The quick-acting valve shall be fully opened and the check valve pressurized with oxygen to a pressure of 2,000 psig. This pressure shall be maintained for 30 seconds and then released through the bleed valve. The valve shall be disconnected and examined for indication of charring or deterioration of any nonmetallic material. The valve shall pass the requirements specified in 3.5.12.

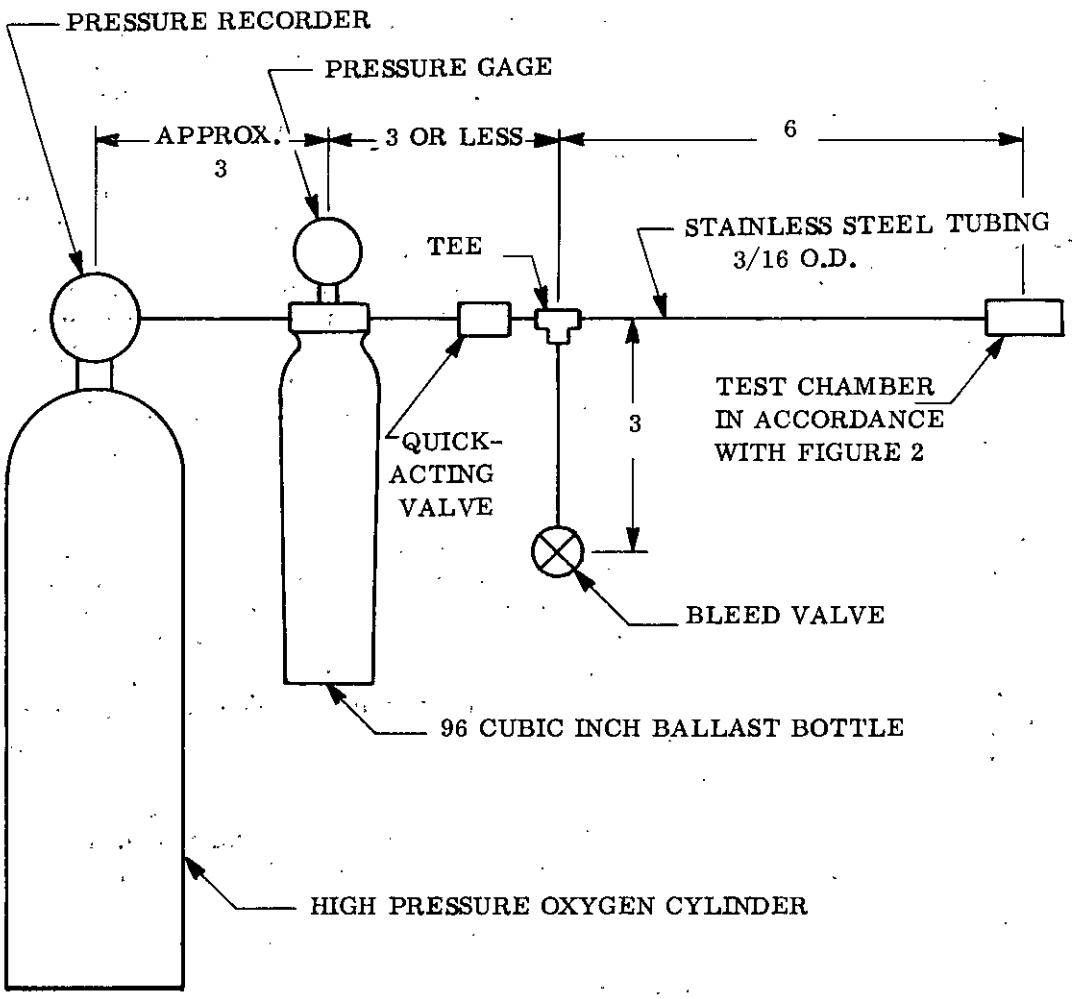
4.7.14 Disassembly and examination - At the conclusion of all the the inspections, the valve shall be disassembled and examined. The valve shall be disassembled and examined after any test in which valve failure indicated faulty construction. The valve shall pass the requirements specified in 3.5.13.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Unless otherwise specified in the contract or order, the valves shall be prepared for delivery in accordance with MIL-STD-794 (see 6.2).

5.1.1 Level A - Each valve shall be preserved in accordance with MIL-P-116, Method 1A13. No contact preservation is required. Each valve shall be sealed within a snug fitting bag of 0.0015 inch gage material conforming to MIL-F-22191, Type III. The sealed valve shall be cushioned within a waterproof fiber can with sealed metal ends. The valve shall be cushioned in such a manner as to prevent damage

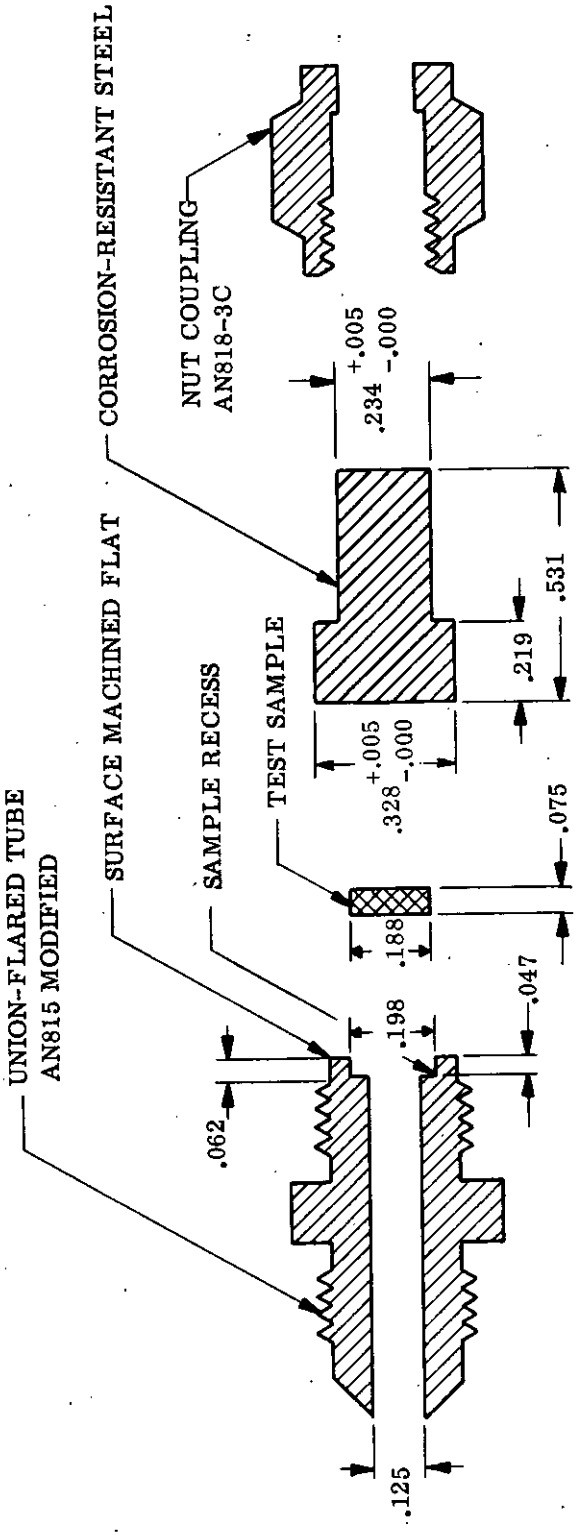
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DIMENSIONS IN INCHES.

FIGURE 1. Oxygen Bomb Equipment

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE, DECIMALS: ± .005

FIGURE 2. Test Chamber

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to itself. Ten unit packages shall be packaged within a fiberboard container conforming to PPP-B-636, Type I, Class 2, Grade 3. The manufacturer's seam shall be adhesive sealed. All seams and joints shall be sealed with tape conforming to PPP-T-60.

5.1.2 Level B - Each valve shall be preserved in accordance with Method 1A13 of MIL-P-116.

5.1.3 Level C - Each valve shall be provided preservation which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the suppliers commercial practice when such meets the requirements of this level.

5.2 Packing - Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A - One hundred valves, packaged as specified in 5.1.1, shall be packed in overseas type exterior containers conforming to PPP-B-591, PPP-B-636, PPP-B-640, Class 2; PPP-B-601 or PPP-B-621 as specified (see 6.2). Unless otherwise specified, PPP-B-601 containers shall be constructed of NN-P-530 plywood conforming to standard grade with exterior glue of U.S. Product Standard PSI-66. This plywood shall have the grade stamp of an approved testing agency and shall be surface treated in accordance with TT-W-572. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitations of the applicable container specification.

5.2.2 Level B - One hundred valves, packaged as specified in 5.1.1, shall be packed in domestic type exterior containers conforming to PPP-B-591, PPP-B-636, PPP-B-640, Class 1; PPP-B-601 or PPP-B-621 as specified (see 6.2). Unless otherwise specified, PPP-B-601 containers shall be constructed of NN-P-530 plywood conforming to standard grade with exterior glue of U.S. Product Standard PSI-66. This plywood shall have the grade stamp of an approved testing agency. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitation of the applicable container specification.

5.2.3 Level C - All packaged valves which require over-packing for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point

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of delivery. Containers shall meet the uniform freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking - Nomenclature tags shall be inserted in each bag showing title, part number or other data as required by procuring activity. In addition, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking - The following precautionary marking shall appear on each package:

"CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN VALVES."

6. NOTES

6.1 Intended use - The check valves are intended for use in high pressure oxygen systems in aircraft.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) AN or MS part number, type and style required (see 1.2 and 3.4).

(c) Whether first article inspection is required and where the samples should be delivered (see 4.4.1.1 and 4.4.1.2).

(d) Selection of applicable levels of preservation, packaging and packing required.

(e) Applicable shipping containers when Level A or Level B packing is specified (see 5.2.1 and 5.2.2).

(f) If plywood in PPP-B-601 or PPP-B-621 for Level A or Level B may be other than specified.

(g) Applicable methods of cleaning and preservation.

(h) Samples subjected to destructive test are not to be considered or shipped as part of the contract or order (see Table II.)

(i) Items of data required (see 6.3).

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6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.4 With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Washington, D.C., 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Department indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4.1 When requested, the manufacturer shall submit engineering drawings and inspection reports in accordance with SD-6, Provisions Governing Qualification.

Custodians:
Navy - AS
Air Force - 71
Army - AV
Review Activities:
AF - (11), (71)
Navy - AS
Army - AV, GL

Preparing Activity:
Navy - AS

(Project No. 1660-0093-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-V-5027D VALVES, CHECK, OXYGEN, HIGH PRESSURE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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